

For deep quantitative research, deep learning, and future-focused stock market modeling, I recommend building mastery in **three layers**:

1. **Foundations** (NumPy, Pandas, Statistics, Visualization)
2. **Machine Learning & Quant Research** (Scikit-Learn, Feature Engineering, Validation)
3. **Deep Learning & Next-Generation Forecasting** (TensorFlow, Keras, Transformers, Time-Series AI)

Top 20 Online Resources

A. Core Python Quantitative Research (1-6)

1. NumPy Official Documentation

The foundation of numerical computing, vectorization, matrix algebra, and quantitative finance operations. [\[numpy.org\]](https://numpy.org)

[NumPy Documentation](https://numpy.org)

Focus on:

- ndarray
- Broadcasting
- Linear Algebra
- Random Sampling
- Statistical Functions

2. Pandas Documentation

The single most important resource for stock research.

https://pandas.pydata.org/docs/user_guide/index.html

Study:

- Time Series
- Resampling
- Window Functions
- Groupby
- MultiIndex
- Merge Operations

3. Seaborn Tutorial

Learn professional statistical visualization. [\[seaborn.pydata.org\]](https://seaborn.pydata.org)

[Seaborn Tutorial](https://seaborn.pydata.org)

Research Applications:

- Correlation Analysis
- Return Distributions
- Regime Visualization
- Pair Relationships

4. Python for Data Analysis (Official Companion)

Created by Wes McKinney (Pandas creator).

<https://wesmckinney.com/book/>

Master:

- Data Wrangling
- Financial Datasets
- Time-Series Analytics

5. Practical Statistics for Data Scientists

<https://github.com/gedeck/practical-statistics-for-data-scientists>

Critical Topics:

- Hypothesis Testing
- Correlation
- Bootstrap
- Regression

6. Quantopian Lectures Archive

Still one of the best free quant resources.

https://www.youtube.com/playlist?list=PLRFLF1OxMm_UL7WUWM31iynp0uL4Hf1nF

Focus:

- Factor Research
- Backtesting
- Portfolio Theory

B. Machine Learning for Quantitative Finance (7-12)

7. Scikit-Learn User Guide

The gold standard for traditional ML. [\[scikit-learn.org\]](https://scikit-learn.org)

[Scikit-Learn User Guide](https://scikit-learn.org)

Master:

- Pipelines
- Feature Selection
- TimeSeriesSplit
- Ensemble Methods
- Clustering

8. Hands-On Machine Learning

Author: Aurélien Géron

<https://github.com/ageron/handson-ml3>

Focus:

- Regression
- Classification
- Ensemble Models
- Neural Networks

9. Kaggle Time Series Course

[Kaggle Time Series Course](#)

Learn:

- Trend Analysis
- Seasonality
- Forecasting
- Feature Engineering

10. Kaggle Feature Engineering Course

[Kaggle Feature Engineering](#)

Very important for stock research.

11. SHAP Documentation

Model explainability.

[SHAP Documentation](#)

Understand:

- Why a prediction happens
- Which features matter
- Market interpretation

12. PyPortfolioOpt

Portfolio optimization research.

[PyPortfolioOpt Documentation](#)

Study:

- Efficient Frontier
- Risk Optimization
- Sharpe Maximization

C. TensorFlow & Keras Deep Learning (13-16)

13. TensorFlow Official Learning Path

[TensorFlow Tutorials](#)

Learn:

- Tensors
- Data Pipelines
- RNNs
- Transformers

14. Keras Documentation

[Keras Documentation](#)

Master:

- Sequential Models
- Functional API
- Custom Layers
- Attention Networks

15. Deep Learning Specialization

Andrew Ng

<https://www.coursera.org/specializations/deep-learning>

Topics:

- Neural Networks
- Optimization
- CNNs
- Sequence Models

16. TensorFlow Time Series Forecasting

[TensorFlow Time Series Guide](#)

Directly relevant for:

- LSTM
- GRU
- Sequence Forecasting

D. Advanced Quant & Future Deep Learning (17-20)

17. Marcos López de Prado Resources

The most influential modern quant researcher.

<https://www.ml4trading.io>

Study:

- Feature Importance
- Labeling Methods
- Financial ML

18. Machine Learning for Trading

<https://github.com/stefan-jansen/machine-learning-for-trading>

Excellent systematic learning path.

Covers:

- Pandas
- NumPy
- Scikit-Learn
- Deep Learning
- Portfolio Construction

19. Hugging Face Time Series Models

Future of forecasting.

[Hugging Face Time Series](#)

Study:

- Transformers
- Foundation Models
- Forecasting LLMs

20. Google Research TimesFM

One of the most important recent forecasting models.

<https://github.com/google-research/timesfm>

Learn:

- Foundation Models for Time Series
- Zero-Shot Forecasting
- Large-Scale Temporal Modeling

Deep Future Learning Roadmap (12 Months)

```
1 Phase 1
2 (NumPy + Pandas)
3   ↓
4 Data Cleaning
5 Rolling Metrics
6 Returns Analysis
7
8 Phase 2
9 (Seaborn)
10  ↓
11 EDA
12 Correlation
13 Market Regimes
14
15 Phase 3
16 (Scikit-Learn)
17  ↓
18 Random Forest
19 XGBoost
20 Feature Importance
21
22 Phase 4
23 (TensorFlow + Keras)
24  ↓
25 Dense Networks
26 LSTM
27 GRU
28
29 Phase 5
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30 (Advanced DL)
31   ↓
32 CNN-LSTM
33 Attention Models
34 Transformers
35
36 Phase 6
37 (Future AI)
38   ↓
39 TimesFM
40 PatchTST
41 Temporal Fusion Transformer
42 Time-Series Foundation Models
43
44 Phase 7
45 (Professional Quant)
46   ↓
47 Portfolio Optimization
48 Factor Investing
49 Risk Modeling
50 Algorithmic Trading
```

My strongest recommendation

For stock research in 2026 and beyond, spend your time in roughly this ratio:

1	Pandas & Feature Engineering	35%
2	Statistics & Research	20%
3	Scikit-Learn	20%
4	TensorFlow/Keras	15%
5	Transformers & Foundation Models	10%

That allocation most closely reflects what successful quantitative researchers actually do, and it will give you a much higher return than focusing mainly on neural networks.